A SMOOTH MORPHISM IS NORMAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5552-5555 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5557-5570, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The characterization of normal morphisms (lemma-normal, same chapter), smoothness implies finite presentation, flatness, and stability of smoothness under base change, and smooth over a field is geometrically normal (Varieties); each is cited with its lemma in the retained span.

Source reference. Original lemma 0391: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5538–5545. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5570–5570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5570–5570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5570–5570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056T: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5570–5570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A smooth morphism is normal.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $f : X \to Y$ be a smooth morphism. As f is locally of finite presentation, see Morphisms, Lemma 01VE the fibres X_y are locally of finite type over a field, hence locally Noetherian. Moreover, f is flat, see Morphisms, Lemma 01VF. Finally, the fibres X_y are smooth over a field (by Morphisms, Lemma 01VB) and hence geometrically normal by Varieties, Lemma 056T. Thus f is normal by Lemma 0391. □